

Adrian Peña

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EDUCATION

University of Texas Rio Grande Valley

M.S. in Computer Science, GPA: 3.75

B.S. in Computer Science, GPA: 3.90

Edinburg, TX

Jan 2025 – Present

Jan 2023 – Dec 2024

WORK EXPERIENCE

Graduate Research Assistant

Edinburg, TX

University of Texas Rio Grande Valley

Jan 2025 – Present

- **Developed** federated learning research by **designing** and implementing an **optimized** PID-based aggregation strategy in **Python**, **running experiments** under varied data and client setups, and **analyzing** robustness against baseline methods.
- **Served** as lead author on a paper accepted to ICICT 2026, a poster accepted to IEEE CCNC 2026, and supporting author on a paper accepted to IEEE ICMLA 2025, all based on distributed federated learning work.

Teaching Assistant & Graduate Teaching Assistant

Edinburg, TX

University of Texas Rio Grande Valley

Aug 2024 – Present

- **Supported** course instruction by grading, answering questions during labs, and explaining core ideas in a way that made students more comfortable with the material.
- **Provided** clear grading feedback on assignments and exams so students could understand mistakes and improve on future work.

Peer-Led Team Learning Leader

Edinburg, TX

University of Texas Rio Grande Valley

Jan 2024 – December 2024

- **Led** and **taught** weekly Peer-Led Team Learning sessions for 30–35 students and **facilitated collaborative problem solving** in data structures and algorithms in **C++**.
- **Guided** students through board-work and group discussions, encouraging them to explain their reasoning and compare different approaches to problems.

Cross-Functional Representative

Mission, TX

H-E-B

Jun 2013 – Dec 2024

- Cross-trained across all departments, **adapting quickly** to varying responsibilities. **Collaborated** with team leads and **communicated clearly** with customers in a dynamic, cross-functional environment.

PROJECTS

Human-AI Email Trust Measurement Study | *Python, SQL, Firebase, React*

Sep 2025 – Present

- **Designed** and **built** a React stimulus app to capture interaction data such as read time, trust ratings, and author-type conditions for Human–AI interaction evaluation.

Lyrics Sentiment Analysis | *Python, VADER*

Nov 2025 – Dec 2025

- **Merged** Spotify metadata with scraped lyrics for the Top 500 tracks, computed VADER sentiment scores, and **applied** PCA and logistic regression to relate sentiment and release year to top-quartile popularity.

PID Control for Federated Learning | *Python, PyTorch, SQL, Flower*

Jan 2025 – Aug 2025

- **Implemented** CNN's in PyTorch on CIFAR-10 and FEMNIST within the Flower using non-IID client data and adversarial clients.
- **Outperformed** standard federated learning baselines, improving final accuracy by **+3–5%** and stabilizing convergence across 30 rounds using an **optimized** PID-controlled aggregation strategy.

Arte Virtual Gallery Usability Study | *Usability Testing*

Jul 2025

- **Led** a think-aloud usability study for a VR gallery and **proposed** actionable fixes aligned with Microsoft HAX guidelines to resolve key navigation and gesture pain points.

Spotify Scheduler | *React, Node.js, Spotify Web API*

Aug 2024 – Dec 2024

- **Built** a full-stack React app that used Spotify OAuth to schedule updates on selected playlists at set times via a Node.js backend and REST API calls to the Spotify Web API.

SKILLS

- **Programming Languages:** Python, SQL, C++, Java, JavaScript, HTML/CSS
- **Frameworks:** PyTorch, Flower, React, Node.js
- **Developer Tools:** Firebase, MySQL, REST APIs, GitHub, GitHub Actions, Jira, Trello, Excel
- **Libraries:** pandas, numpy, matplotlib, scikit-learn, VADER